POCO F4 | ANURAG

0108IT241015

Paper Code: PYB-101
Examination –May- 2025
B.Tech. II Sem :
Subject Name: Applied Physics

Time : 3 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a(i) and a(ii) Objective type/Fill in the blanks/true& false.
Part (b) – Three/four sentences based answer question (Maximum 50words)
Part (c) – Short Descriptive Ans./Numerical/ algorithm /programs based Question. (Maximum 100 words)
Part (d) – Long Answer /Shorts -notes /Descriptive Ans. based Question (Maximum 400 words)
Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)- I	If the material particle moves with kinetic energy $E = ½ mv^2$, then de-Broglie wavelength can be written as (i) $\lambda = \sqrt{2mE}/h$ (ii) $\lambda = hm/\sqrt{2E}$ (iii) $\lambda = h/\sqrt{2mE}$ (iv)None of them	01	2	1
	(a) -II	A wave function that satisfied the condition $\int \psi \psi^* d\tau = 0$ is known as normalization. (True/ False)	01	2	1
	(b)	What is concept of the wave function ψ?	02	3	1
	(c)	Write one experimental proof of the Heisenberg Uncertainty Principle.	03	2	1
		OR			
		X- ray of the wavelength $0.2A^0$ are scatted by the electron in block of carbon through 45^0. Find the wavelength of scattered photon.	03	3	1
	(d)	Derive the relation between the group velocity and particle velocity.	05	4	1
		OR			
		Derive Schrödinger time- independent equation for matter wave.	05	4	1
Q.2	(a)-I	Which of the following is not a laser property? (i) Highly monochromatic (ii) High directionality (iii) Very narrow band width (iv) Highly divergent	01	2	2
	(a)-II	Holography is an interferences based technique (True/ False).	01	1	2
	(b)	Define V-number for a SI fibre.	02	2	2
	(c)	Explain Stimulated Emission process in Lasers.	03	4	2
		OR			
		Calculate the acceptance angle for an optical fiber having refractive index of the core cladding materials are 1.45 and 1.40 respectively.	03	3	2
	(d)	Discuss the construction and working of He-Ne laser.	05	4	2
		OR			
		Derive the expression for Numerical aperture of the step index optical fiber.	05	4	2

Write a program in C to calculate the factorial of a given number using a for loop. Explain the logic behind your code.

OR

What is a union in C? How does it differ from a structure in terms of memory allocation? Explain the declaration and usage of a union with an example. 05 | 5 | 2

Q (a)-I Which of the following best describes the concept of Blockchain? 01 | 1 | 3
a) A centralized database where information is stored in blocks
b) A decentralized ledger of transactions, secured using cryptography
c) A system that only handles cryptocurrency transactions
d) A method for encrypting data in a file system

(a)-II In an operating system, which of the following is responsible for managing hardware resources and ensuring that processes are executed properly? 01 | 2 | 3
a) User Interface b) File Management System
c) Kernel d) Command Interpreter

(b) Explain the concept of blockchain. How does it provide security in transactions? 02 | 2 | 3

(c) What is machine learning, and how does it differ from traditional programming? 03 | 3 | 3

OR

Define cybersecurity.Describe the term encryption. 03 | 3 | 3

(d) Describe the role of an operating system (OS) in a computer system. Explain its functions, including process management, memory management, and file system management, and how it interacts with hardware. 05 | 3 | 3

OR

Explain the fundamental components of computer networking. What are the various types of network topologies, and how do they function? 05 | 5 | 3

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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	(d)	Write a program in C to calculate the factorial of a given number using a `for` loop. Explain the logic behind your code.	05	5	2
		OR			
		What are data types in C? Explain the difference between primitive and user-defined data types with examples.	05	5	2
Q.3	(a)-I	In C, what does the return statement do in a function? a) It ends the program. b) It terminates the function and optionally returns a value to the calling function. c) It pauses the function. d) It starts the function.	01	1	3
	(a)-II	Which type of recursion is efficient in terms of memory usage because it does not involve multiple function calls? a) Tail recursion b) Tree recursion c) Direct recursion d) Indirect recursion	01	1	3
	(b)	What is the purpose of the `return` statement in C functions?	02	2	3
	(c)	What is recursion? Compare recursion with iteration.	03	3	3
		OR			
		Explain the different types of recursion with examples.	03	3	3
	(d)	Discuss the use of arrays in modular programming. Write a function to find the largest element in a one-dimensional array and explain the modular approach used.	05	5	3
		OR			
		Differentiate between call by value and call by reference in C. Write a program to swap two numbers using both methods.	05	5	3
Q.4	(a)-I	Which of the following is used to open a file in C? a) open() b) read() c) fopen() d) fileopen()	01	1	2
	(a)-II	In C, what is the size of a union with two members: int x; and char y;? a) 2 bytes b) 4 bytes c) 1 byte d) 8 bytes	01	1	2
	(b)	What is an enumerated data type in C?	02	2	2
	(c)	Discuss the purpose and usage of the `#define` and `#include` preprocessor directives in C. How do they improve code maintainability?	03	3	2
		OR			
		Explain the concept of self-referential structures with the help of example.	03	3	2
	(d)	Explain the concept of a structure in C. How do you declare a structure and access its members using the membership operator? Explain with help of example.	05	5	2
		OR			